

AUTOMOTIVE ELECTRONICS

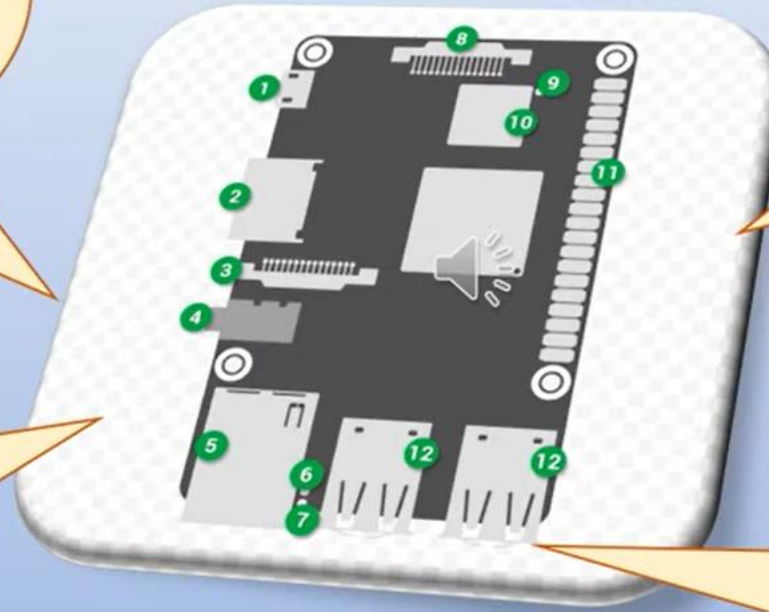
Comprehensive System

Control

Request

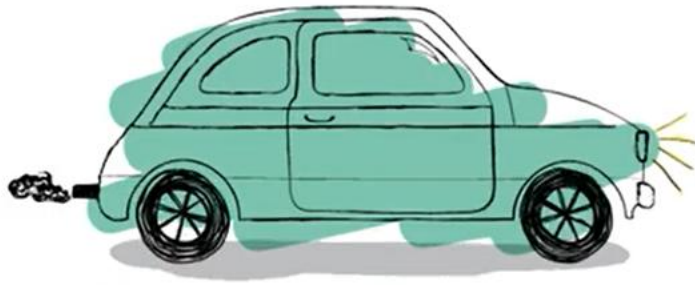
Monitoring

Process



Why Embedded In Automotive

➤ Sustainable Mobility



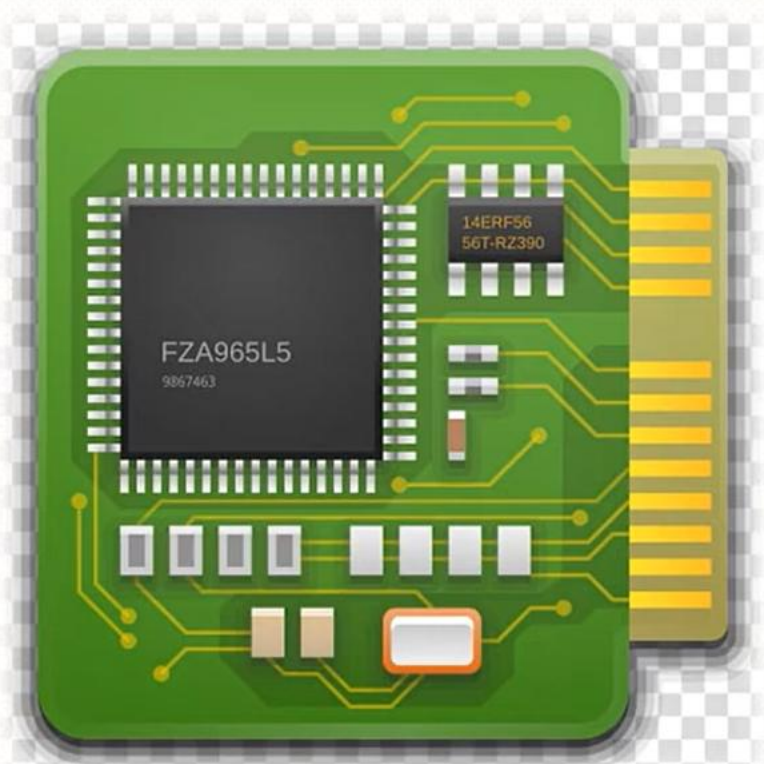
➤ Comfort Mobility



➤ Safety Mobility

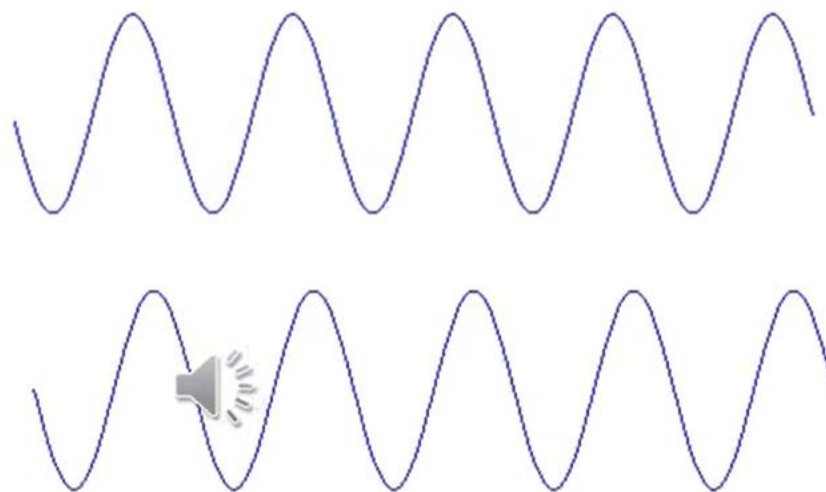


Electronic Control Unit



ECU

Object Sensing



Object

Output of the ECU



Warning to User



ELECTRONICS IN THE AUTOMOBILE

Today upto 80 Microcontrollers and more than 100 electric motors together with voltage regulators, power MOSFETs, relays, sensors, capacitors and inductances, are used in applications such as

- ❖ **BRAKING**

- ❖ Anti-lock Braking System (ABS)
- ❖ Anti-Slip Regulation(ASR)
- ❖ Electronic Stability Program(ESP), etc.,

- ❖ **STEERING**

- ❖ Electro Hydraulic Power Steering (EHPS)
- ❖ Electronic Power Steering (EPS)
- ❖ Rear Steering
- ❖ Fault-Tolerant Steering, etc.,

ELECTRONICS IN THE AUTOMOBILE

❖ SUSPENSION

- ❖ Load Leveling
- ❖ Dynamic Leveling
- ❖ Semi-active Damping
- ❖ Active Suspension, etc.,

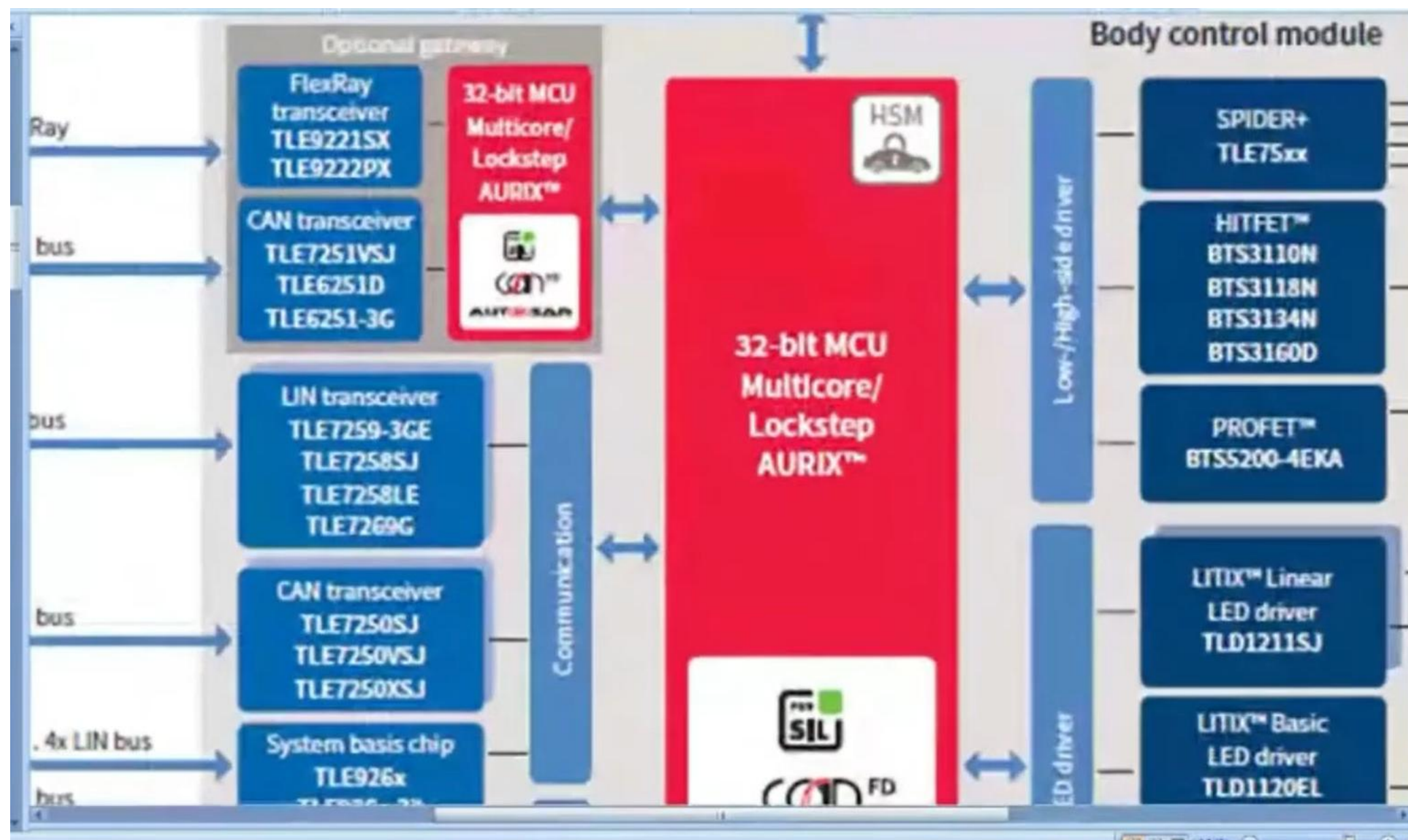
❖ SAFETY MANAGER

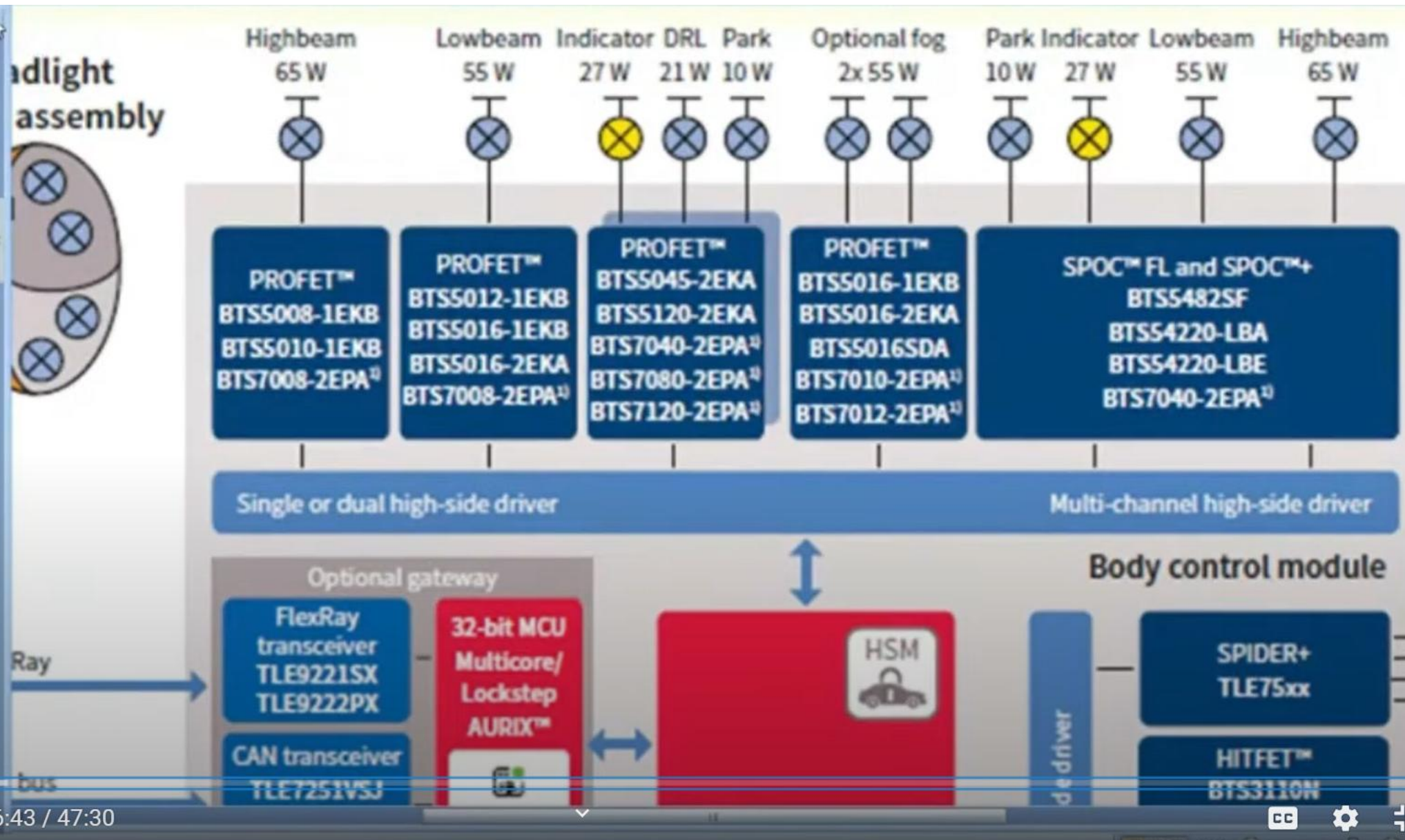
- ❖ Tire Pressure Monitoring System (TPMS)
- ❖ Passenger Detection
- ❖ Seat Belt
- ❖ Air-bag
- ❖ Secure Data Recording, etc.,

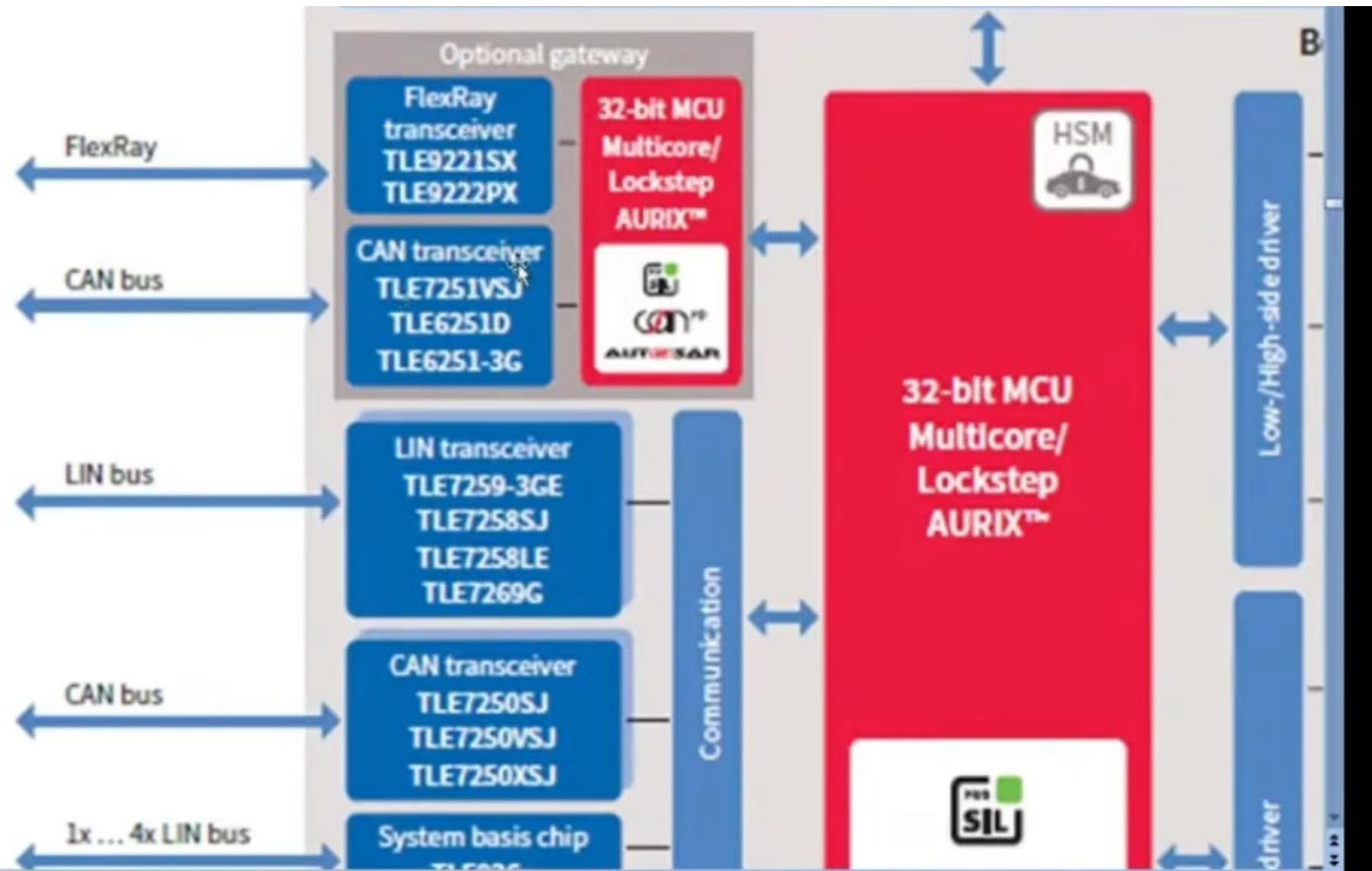
ELECTRONICS IN THE AUTOMOBILE

❖ DRIVER ASSISTANCE

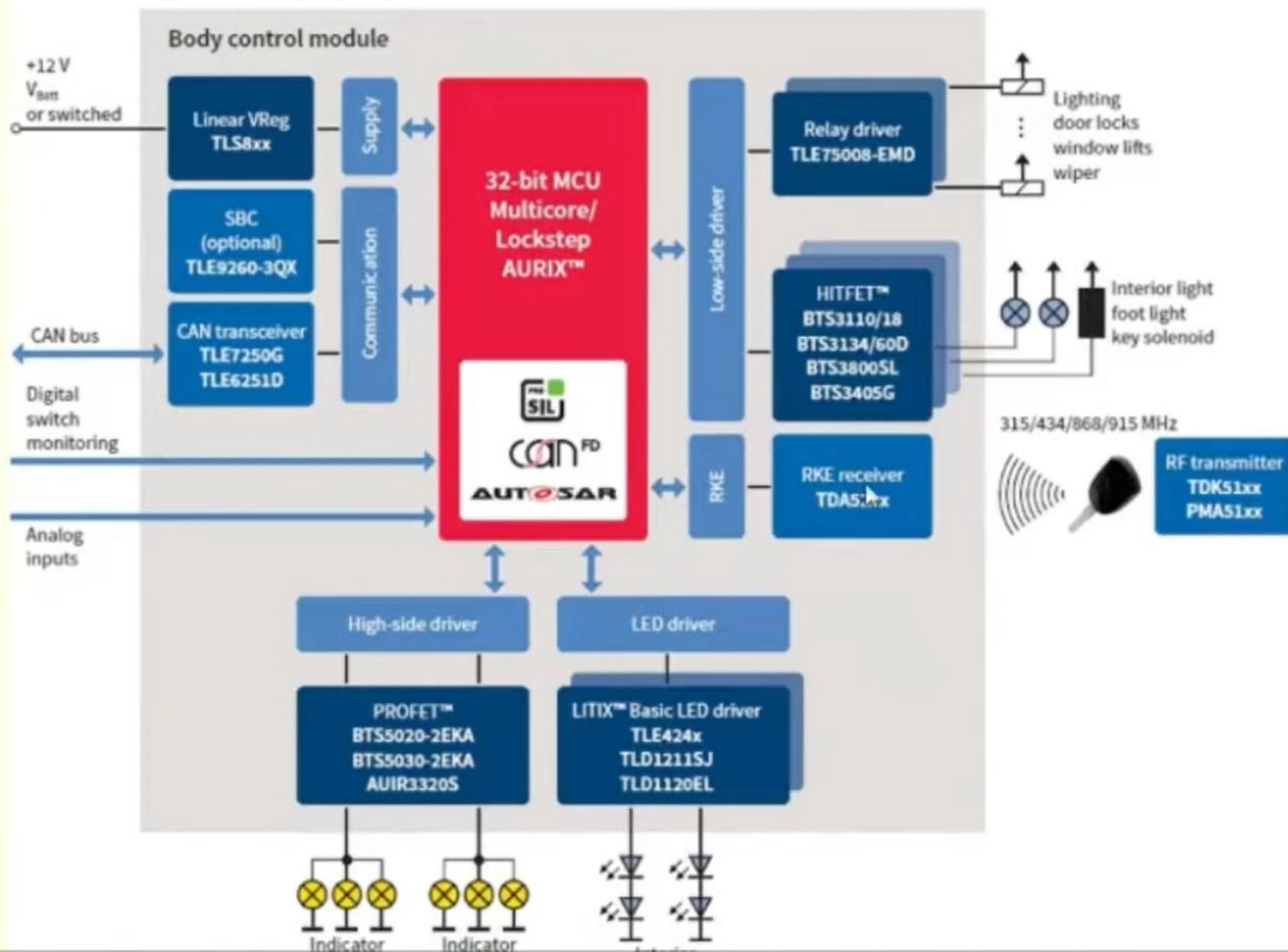
- ❖ Cruise Control
- ❖ Active Cruise Control 1st Generation
- ❖ Active Cruise Control 2nd Generation
- ❖ Lane Departure
- ❖ Vision Enhance
- ❖ Parking Assistant
- ❖ Blind Spot, etc.,



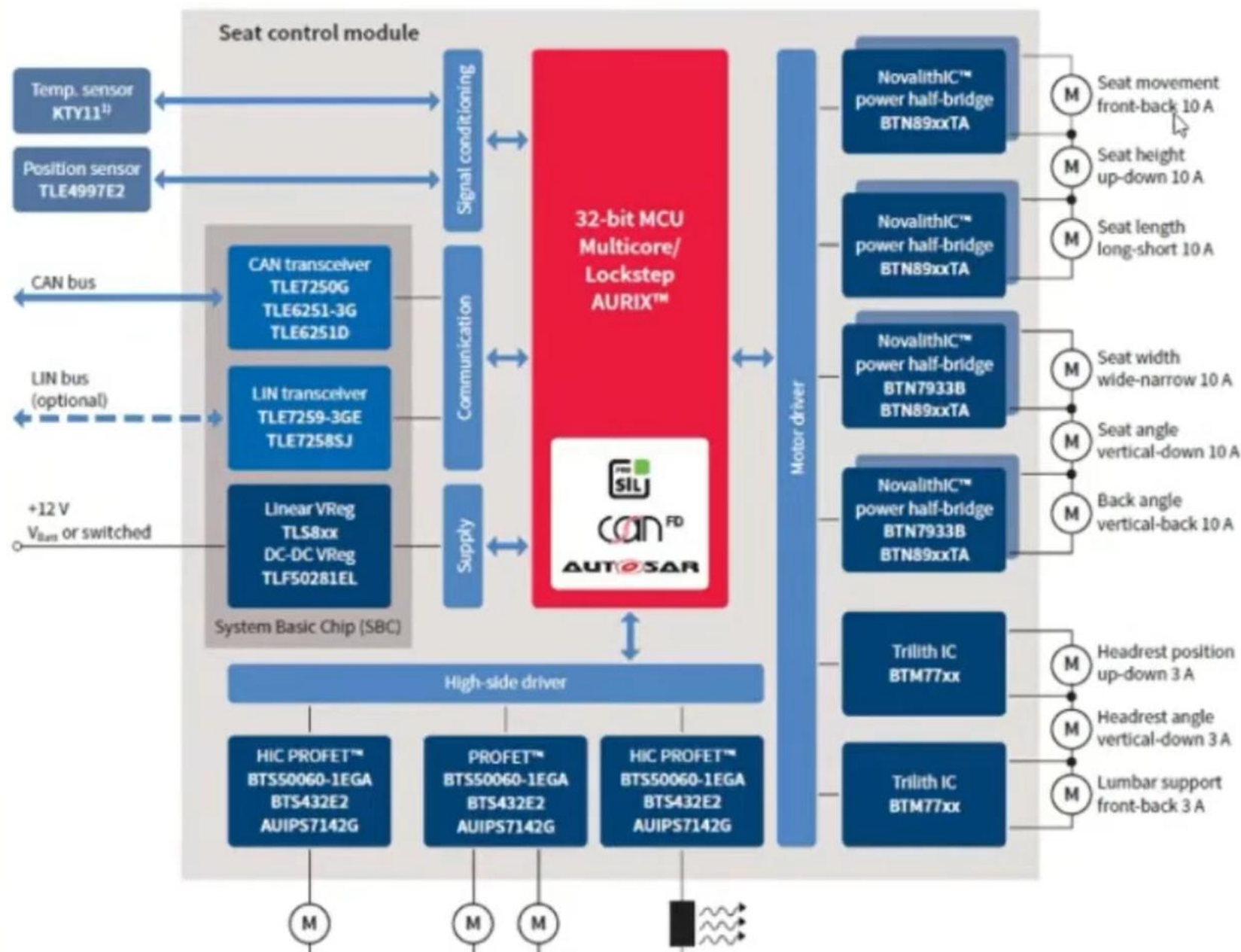




LOW COST BODY CONTROL MODULE



INFINEON's SEAT CONTROL MODULE



SMART WINDOW LIFT / ROOF CONTROL MODULE

